MoLIC-XAI

Semiotic Engineering

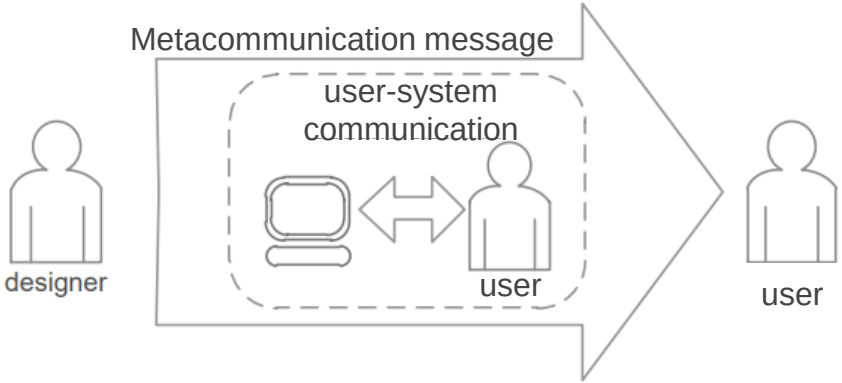
A communication-centered rather than user-centered HCI theory [1]

- General lens: Human-Computer Interaction is a particular case of **computer-mediated human communication**
- Focuses on the communication between users, designers, and systems:
  - Direct communication: User-system communication
  - **Metacommunication**: Communication from the designer (producer) to the user (consumer) mediated by the system via an interface (proxy)
- Computing systems are **metacommunication artifacts**: They communicate a message from the designer to the user regarding the user-system communication
- Tools: **MoLIC-XAI**

Semiotic Engineering

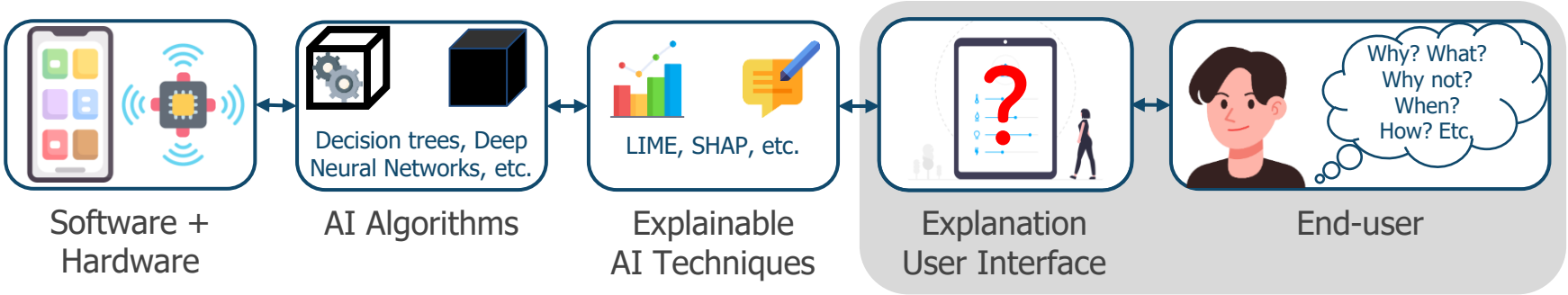
Metacommunication template [1, 2]

Here is my understanding of **who** you are, what I've learned you want or **need** to do, in which **preferred ways**, and **why**. This is the system that I have therefore **designed** for you, and this is the way you can or should **use** it in order to fulfill a range of **purposes** that fall within this vision



Explainable AI (XAI)

Explainable AI (XAI) system components [3]



Appropriate reliance: In decision-making processes end-users should know **when to rely** on AI's advice and **when not to rely** on it, avoiding over- and under-reliance

Explanation Types [4]

Local explanations: Justify AI reasoning at the prediction level by quantifying feature contributions or generating local rules

Global explanations: Provide insights into the overall model logic, such as feature importance rankings or decision trees

Example-based explanations: Justify AI outputs by presenting similar cases from the dataset

Counterfactual explanations: Answer “what-if” questions by showing how modifying an input would change the AI's decision

Explanation Types

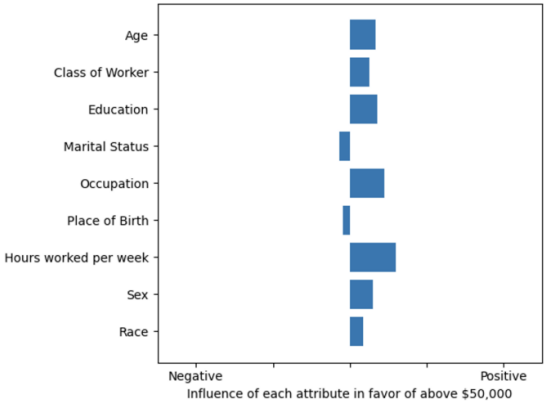
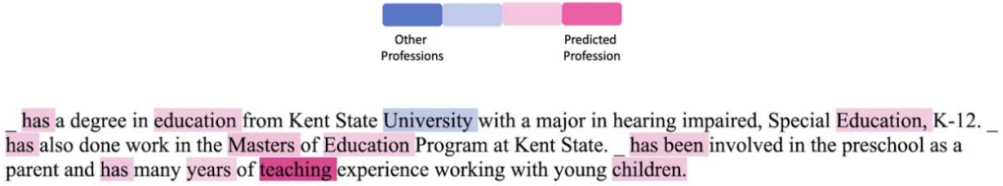
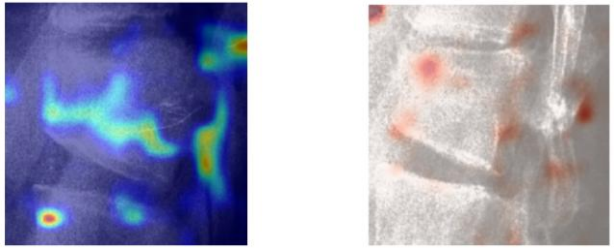


Chart – Feature contribution [5]



Textual + Color scale [5]



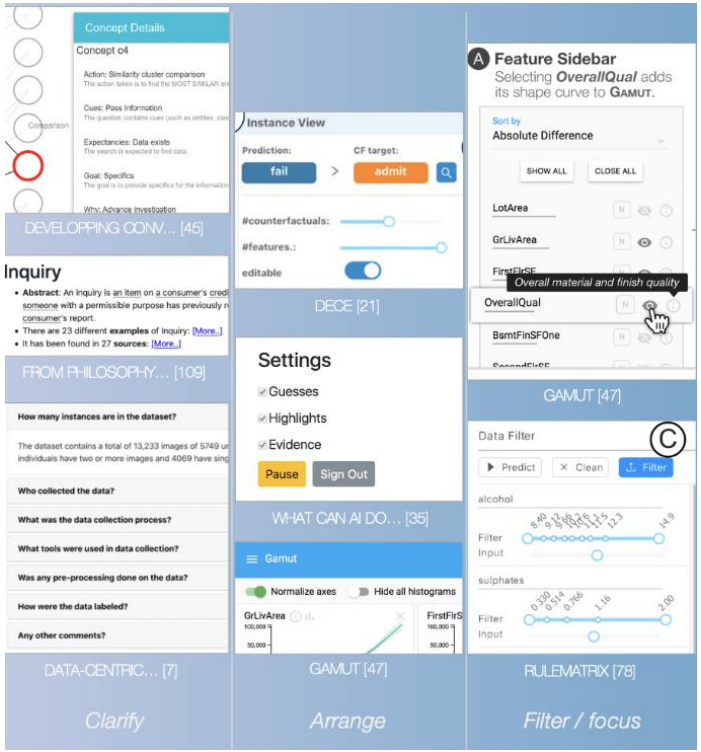
Saliency-based highlights [6]

Interactive Explanations

Taxonomy of interactivity techniques in XAI [7]

Selective Explanations – Tasks:

- ❑ Clarify: Give additional information/explanations on demand
- ❑ Arrange: Choose and organize the explanation type(s), parameters and visual representation(s)
- ❑ Filter/Focus: Filter the explanation according to an input/input metric

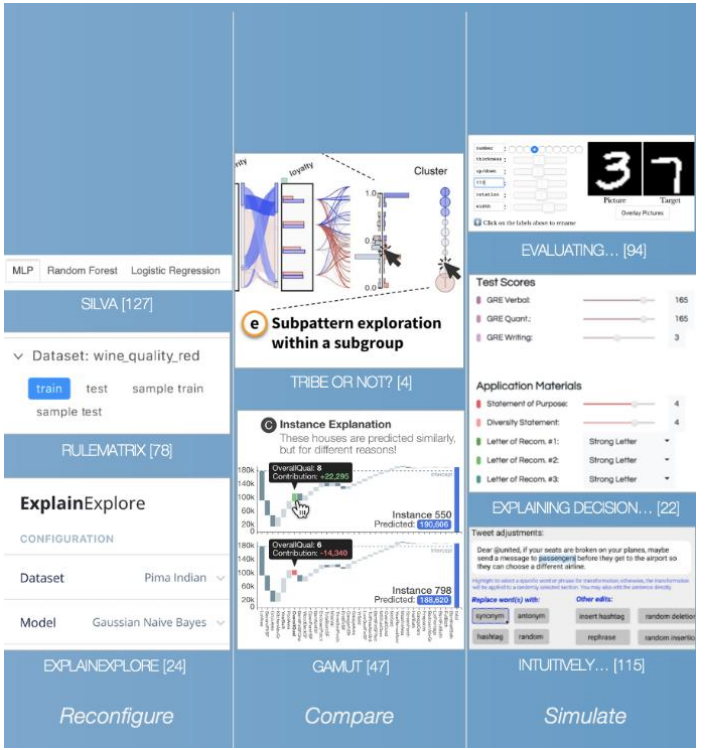


Interactive Explanations

Taxonomy of interactivity techniques in XAI [7]

Mutable Explanations – Tasks:

- ❑ Reconfigure: Change the dataset, the AI model, AI model parameters and show me the corresponding prediction and explanations
- ❑ Simulate: Change the inputs, the output or the dataset distribution and show me the corresponding prediction and explanations
- ❑ Compare: Show me explanations of related or selected data inputs or outputs

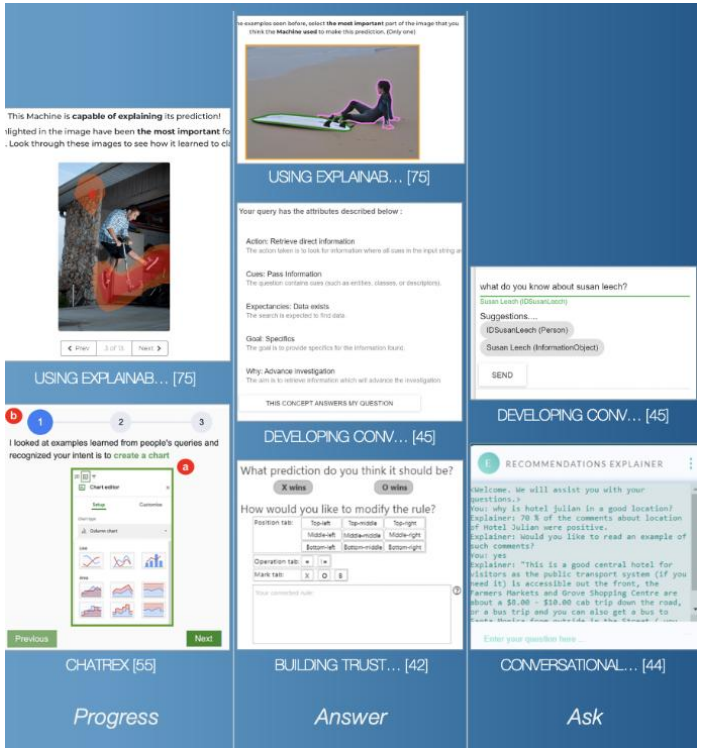


Interactive Explanations

Taxonomy of interactivity techniques in XAI [7]

Dialogic Explanations – Tasks:

- ❑ Progress: Guide user through an explanation sequence
- ❑ Answer: Give feedback, edit explanation components
- ❑ Ask: Iterative questions and receive answers following a dialogue structure



MoLIC-XAI

Modeling Language for Interaction as Conversation [1, 8]

- Represents HCI as a set of dialogues between the user and the designer's proxy
- The proxy communicate: What the system **did** (or didn't do), what it is **doing** (or not doing), what it **allows** or **prohibits** users from doing, **how** and **why**
- Magnifies designer's compreheension regarding the problem being solved and the artifact being designed
- **+XAI:** Elements to represent content generated by AI and interactive explanations

MoLIC-XAI Notation

Starting and Ending

Conversation starting point

Conversation ending point

MoLIC-XAI Notation

Scenes

Login

inform credentials {
d+u: username, password}

Normal scene

View Loan Packages

view available packages{
d: list (available packages)}

Scene + list function

Confirmation

d: Are you sure to delete the current loan proposal?

Alert scene

View Results

view explanation{
d: loan approved of 50% higher than budget y
d: confidence level of 95%
d: feature contribution for my recommendation}

AI-generated content scene

Confirmation

inform feedback{
d+u: list (feedback options)?}

AI-generated alert scene

View Additional Explanation

if addition explanation requested{
d: textual explanation for my prediction}

Selective explanation scene

View Similar Projects

view similar projects{
d: projects A, B, and C have a similar prediction as project Y
d: confidence level for each project
d: feature contribution for each prediction}

Mutable explanation scene

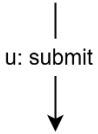
Provide Feedback

inform feedback{
d+u: list (feedback options)?}

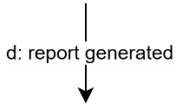
Dialogic explanation scene

MoLIC-XAI Notation

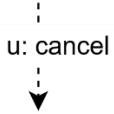
Utterances



User’s utterance



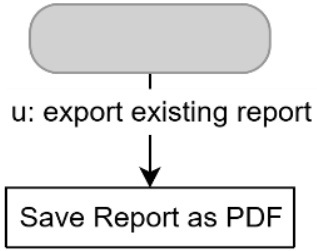
Designer’s uterrance



User’s repair utterance

MoLIC-XAI Notation

Global topic shift



MoLIC-XAI Notation

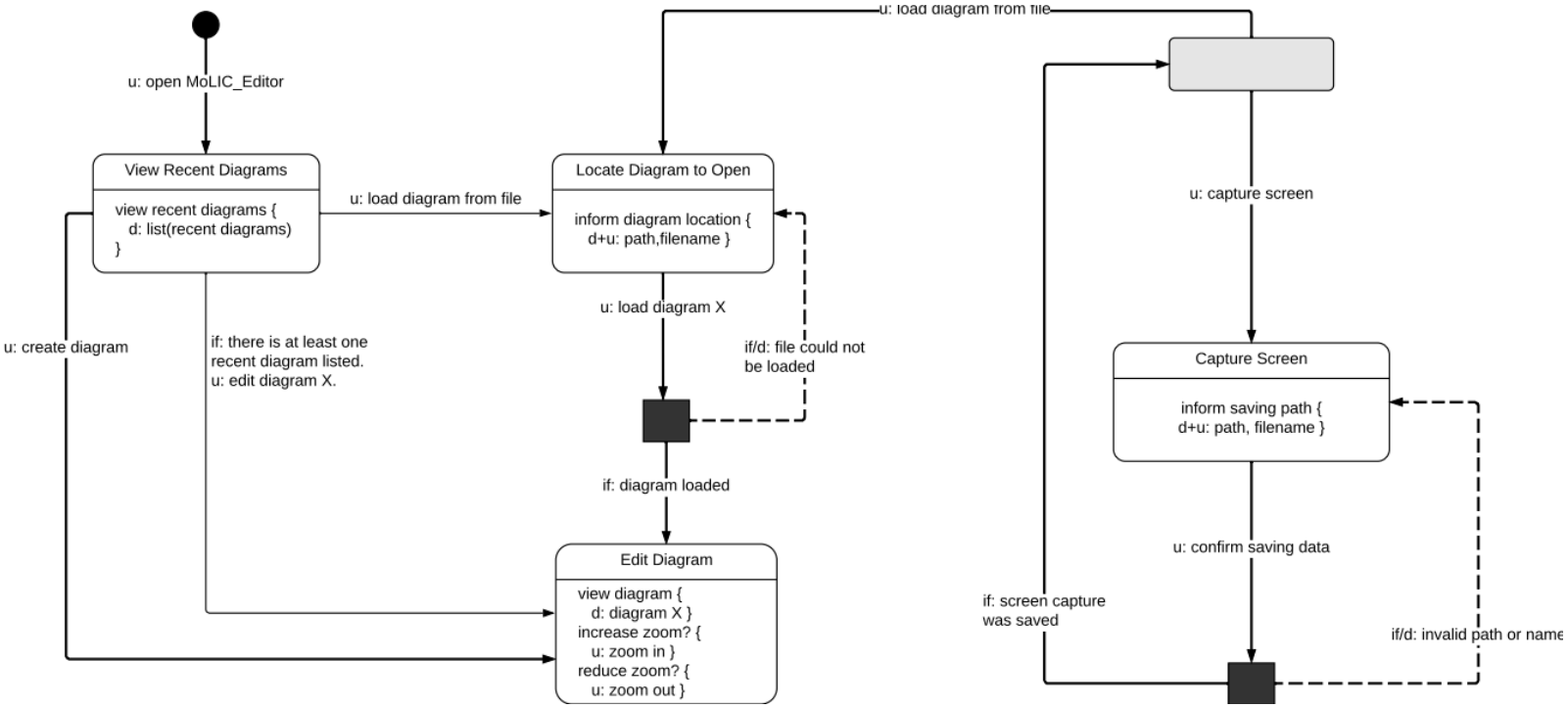
Processing

System process

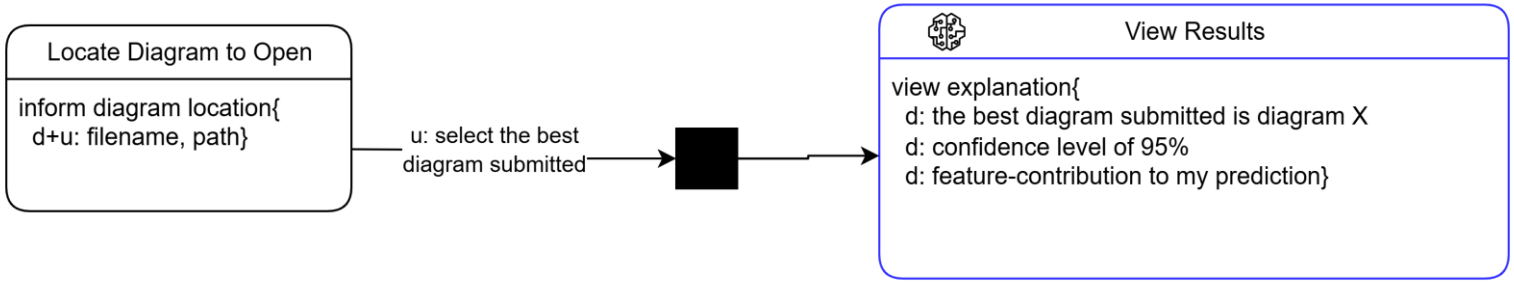
d: % concluded

System process with designer’s feedback

MoLIC-XAI Example [8]



MoLIC-XAI Example



References

[1] Barbosa, S. D. J., Silva, B. D., Silveira, M. S., Gasparini, I., Darin, T., & Barbosa, G. D. J. (2021). **Interação humano-computador e experiência do usuario**. Auto publicação.

[2] de Souza, C. S. (2005). **The Semiotic Engineering of Human-Computer Interaction** (Acting with Technology). The MIT Press.

[3] Gunning, D., Vorm, E., Wang, Y., & Turek, M. (2021). **DARPA's explainable AI (XAI) program: A retrospective**. Authorea Preprints.

[4] Liao, Q. V., & Varshney, K. R. (2021). **Human-centered explainable ai (xai): From algorithms to user experiences**. arXiv preprint arXiv:2110.10790.

[5] Chen, V., Liao, Q. V., Wortman Vaughan, J., & Bansal, G. (2023). **Understanding the role of human intuition on reliance in human-AI decision-making with explanations**. Proceedings of the ACM on Human-computer Interaction, 7(CSCW2), 1-32.

[6] Natali, C., Famiglini, L., Campagner, A., La Maida, G. A., Gallazzi, E., & Cabitza, F. (2023). **Color shadows 2: Assessing the impact of xai on diagnostic decision-making**. In World Conference on Explainable Artificial Intelligence (pp. 618-629). Cham: Springer Nature Switzerland.

[7] Bertrand, A., Viard, T., Belloum, R., Eagan, J. R., & Maxwell, W. (2023). **On selective, mutable and dialogic XAI: A review of what users say about different types of interactive explanations**. In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (pp. 1-21).

[8] Barbosa, S. D. J. & Paula, M. G. (2003). **Designing and Evaluating Interaction as Conversation: A Modeling Language Based on Semiotic Engineering**. In Interactive Systems. Design, Specification, and Verification, DSV-IS 2003, Lecture Notes in Computer Science, pages 16--33, Berlin, Heidelberg. Springer.